

India's Digital Revolution

IT Services, UPI, and the Software Superpower That Built Itself

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Executive Summary

India's technology sector is undergoing rapid structural development across digital payments infrastructure, semiconductor manufacturing, AI data centre investment, and telecom connectivity. Tata Electronics has secured Intel as a prospective customer for its upcoming chip facilities as part of a \$14 billion semiconductor foray CNA[2025-12-08]. Reliance Industries' joint venture has committed \$11 billion over five years to develop 1 gigawatt of AI data capacity in Andhra Pradesh CNA[2025-11-26], while Adani Enterprises has announced a \$100 billion investment in AI-ready data centres by 2035 CNA[2026-02-17]. On the payments side, the Reserve Bank of India and the European Central Bank have agreed to begin an initial phase of interlinking domestic payment systems CNA[2025-11-21], extending the Unified Payments Interface's international reach.

Part I — Digital Payments and the UPI Network

UPI: Cross-Border Expansion

India's Unified Payments Interface (UPI) has become a central pillar of India's cross-border digital payments strategy. India's UPI is one of five national instant payment systems — alongside Singapore's PayNow, Malaysia's DuitNow, Thailand's PromptPay, and the Philippines' InstaPay — participating in Project Nexus, a collaboration between their central banks to create a global network for instant cross-border payments CNA[2025-01-19].

Singapore's PayNow and India's UPI were linked by July 2022 to allow instant, low-cost fund transfers directly between bank accounts in the two countries CNA[2021-09-14].

Most recently, the Reserve Bank of India and the European Central Bank agreed to start the initial phase of interlinking their domestic payments systems, the RBI announced CNA[2025-11-21]. This would extend UPI connectivity toward the European payments landscape.

Biometric Authentication for UPI

India rolled out biometric authentication for UPI transactions starting October 2025, allowing users to approve payments using facial recognition and fingerprints CNA[2025-10-07]. The move extends the digital identity layer into the payments flow and is intended to reduce friction for users who find PIN-based authentication cumbersome.

Central Bank Stance on Crypto and Stablecoins

Reserve Bank of India Governor Sanjay Malhotra stated in November 2025 that the central bank is adopting a cautious approach towards cryptocurrencies and stablecoins CNA[2025-11-20]. The RBI's

conservatism on crypto stands in contrast to India's aggressive embrace of state-backed digital payment rails.

Part II — Semiconductors: The Upstream Ambition

Tata Electronics and Intel

India's Tata Electronics secured Intel as a prospective customer for its upcoming chip facilities — a development described as potentially signalling the US chipmaker's confidence in India's manufacturing ambitions. The electronics-manufacturing arm of the 156-year-old Tata conglomerate is pursuing a \$14 billion chip foray CNA[2025-12-08].

Part III — AI Data Centre Investment

Reliance Industries Joint Venture

A Reliance Industries joint venture will invest \$11 billion over five years to develop 1 gigawatt of AI data capacity in the southern Indian state of Andhra Pradesh CNA[2025-11-26]. The scale of the commitment — 1 GW in a single state — reflects the intensity of the race to build AI-ready computing infrastructure in India.

Tata Consultancy Services and TPG

India's Tata Consultancy Services and private equity firm TPG will form a joint venture to develop AI data centres, with both partners set to invest \$2 billion CNA[2025-11-20].

Adani Enterprises

Adani Enterprises announced in February 2026 that the group will invest \$100 billion to build renewable energy-powered AI-ready data centres by 2035, aiming to create what it described as the world's largest integrated data centre platform CNA[2026-02-17].

Part IV — Telecom and Connectivity

Reliance Jio and AI Access

Google will offer 18 months of free access to its Gemini AI service for all 505 million telecom users of India's Reliance Jio, a tie-up that follows similar partnerships between AI providers and large telecom operators to accelerate AI adoption in India CNA[2025-10-30].

5G Rollout

India launched 5G network trials in October 2022, with faster download speeds and higher bandwidth positioned as essential for adopting emerging technologies including artificial intelligence and robotics. Reliance Industries Chairman Mukesh Ambani and his son Akash Ambani, Chairman of Jio, attended the launch of 5G services in New Delhi CNA[2022-10-10].

Conclusion

The evidence base assembled here covers four intersecting dimensions of India's technology development: digital payments internationalisation (UPI linkages with Singapore, ASEAN, and now the EU), semiconductor manufacturing ambition anchored by Tata's fabrication foray with Intel as a prospective customer, AI data centre investment at a scale that places India among the most active

markets globally, and telecom infrastructure that now touches hundreds of millions of Jio subscribers with AI-augmented services. The pace of investment commitments suggests a structural shift in India's technology ambitions from software services toward hardware and infrastructure — though execution against these commitments remains to be demonstrated.

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